Exp 1 · Logic gates, and NAND/NOR as universal gates

Manual pp. 1–4 · ICs 7400, 7402, 7404, 7408, 7432, 7486 · trainer kit

1 Gates & universality → 2 Adders/subtractors → 3 NAND/NOR design → 4 Code converters

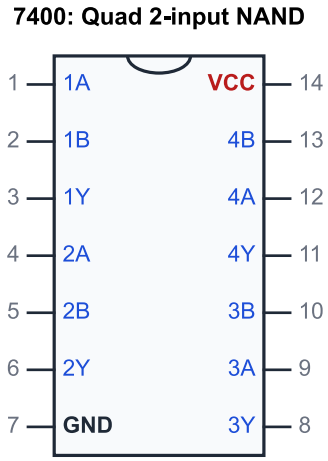
MANUAL EXP 1, AIM

Verify the truth tables of AND, OR, NOT, NAND, NOR and XOR; realise NOT, AND and OR using only NAND gates, then only NOR gates. Viva form: "Why are NAND and NOR called universal? Show OR from NANDs only."

GUESS FIRST

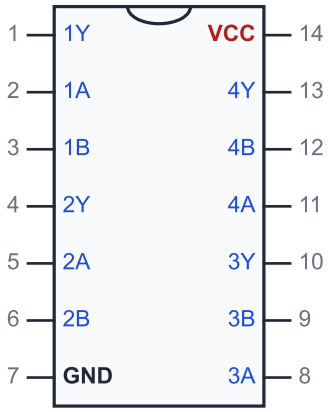
How many 2-input NANDs make an OR? Sketch it before reading on.

The six ICs



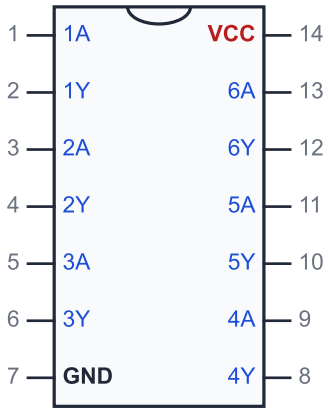
7408 (AND), 7432 (OR) and 7486 (XOR) use exactly this layout. Only the gate inside changes.

7402: Quad 2-input NOR



The 7402 is different: each output comes FIRST (pins 1, 4, 10, 13).

7404: Hex inverter



Six inverters.

Testing a gate IC (e.g. the 7408): connect A → pin 1, B → pin 2, LED ← pin 3, VCC 14, GND 7. Walk 00, 01, 10, 11. Repeat for gates 2–4 if you have time: a chip can have one dead gate.

A B	AND	OR	NAND	NOR	XOR	NOT A
0 0	0	0	1	1	0	1
0 1	0	1	1	0	1	1
1 0	0	1	1	0	1	0
1 1	1	1	0	0	0	0

Universality

A gate is **universal** if NOT, AND and OR (and so any function) can be built from it alone.

Make	From NAND (7400)	From NOR (7402)
NOT A	NAND(A, A)	NOR(A, A)
AND	NAND, then a NAND-inverter: ((AB)')'	invert both inputs, then NOR: (A' + B')' = AB (De Morgan)
OR	invert both inputs, then NAND: (A'B')' = A + B (De Morgan)	NOR, then a NOR-inverter: ((A + B)')'
gates needed	NOT 1, AND 2, OR 3	NOT 1, OR 2, AND 3

Wiring (the manual's pin numbers)

Circuit	Wires
NOT (NAND)	A → 7400 pins 1 and 2; Y = pin 3
AND (NAND)	A → 1, B → 2; pin 3 → pins 4 and 5; Y = pin 6
OR (NAND)	A → 1 and 2 (pin 3 = A'); B → 4 and 5 (pin 6 = B'); pin 3 → 9, pin 6 → 10; Y = pin 8
NOT (NOR)	A → 7402 pins 2 and 3; Y = pin 1
OR (NOR)	A → 2, B → 3; pin 1 → pins 5 and 6; Y = pin 4
AND (NOR)	A → 2 and 3 (pin 1 = A'); B → 5 and 6 (pin 4 = B'); pin 1 → 8, pin 4 → 9; Y = pin 10

WRITE IN YOUR RECORD

Result: the truth tables of AND, OR, NOT, NAND, NOR and XOR were verified. NOT, AND and OR were realised using only NAND gates (1, 2, 3 gates) and using only NOR gates (1, 3, 2 gates), and their truth tables verified.

Theory for the record: logic gates implement Boolean operations: AND (Y = AB), OR (Y = A + B), NOT (Y = A'), NAND (Y = (AB)'), NOR (Y = (A + B)'), XOR (Y = AB' + A'B). TTL ICs of the 74 series contain several gates in a 14-pin DIP, with VCC on pin 14 and GND on pin 7. NAND and NOR are universal gates: NOT, AND and OR, and therefore any Boolean function, can be built using only one of them. By De Morgan's theorems, (A'B')' = A + B and (A' + B')' = AB.

CHECK 1

(a) Make XOR from 2-input NANDs only: how many gates? (b) Why does a floating input on a 7402 NOR give a stuck-at-0 output? (c) Which pin is the output of gate 3 on the 7400, and on the 7402?

Answers

Opening question. Each of them alone can make NOT, AND and OR, and so any function. OR = NAND(NAND(A,A), NAND(B,B)): **3 NANDs**.

Check 1. (a) **4 NANDs**: n1 = (AB)', then A $\oplus$ B = NAND(NAND(A, n1), NAND(B, n1)). (b) A floating TTL input behaves as logic 1, and a NOR with any input at 1 outputs 0. (c) 7400 gate 3: inputs 9, 10 → output **8**. 7402 gate 3: inputs 8, 9 → output **10**.